

Choosing a Motherboard (Mobo)

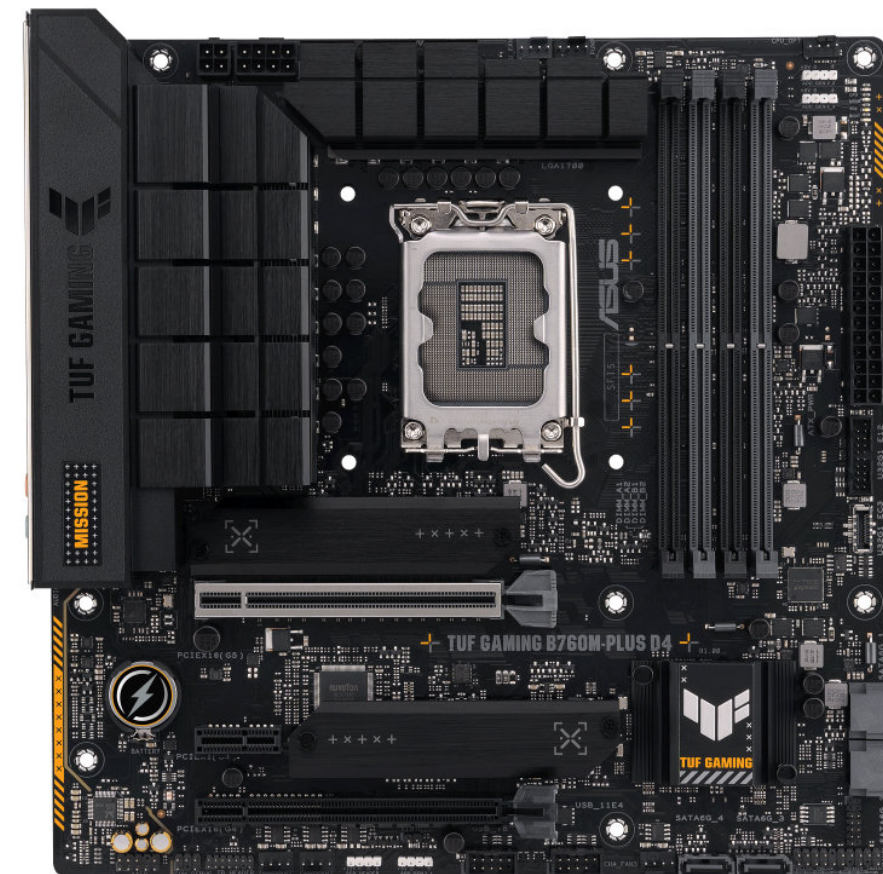
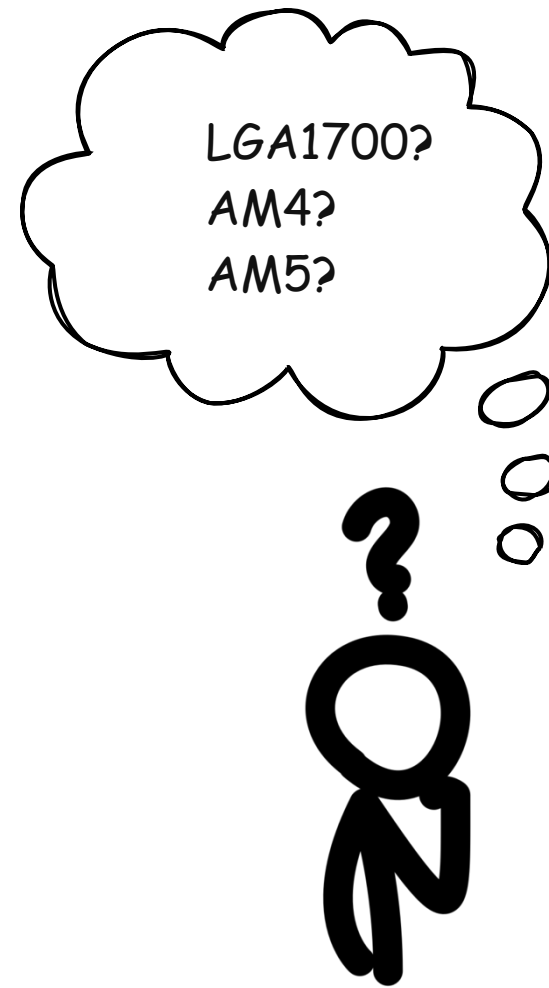
ASRock

ASUS®

 *msi*

NZXT

GIGABYTE™



Market Overview

Most manufacturers have ranges of motherboard aimed at different users, budget, gamer, WFH pro, creator etc...



- **ROG** - High-end gaming (\$\$\$\$\$)
- **ProArt** - Creativity professionals (\$\$\$\$)
- **TUF** - Mid-level gaming (\$\$\$\$)
- **PRIME** - All rounders, entry level (\$\$)
- **Workstation** - productivity and business (\$\$)



- **AORUS** - high-end gaming motherboards (\$\$\$\$\$)
- **AERO** - designed for content creators (\$\$\$\$)
- **Gigabyte Gaming** - mid-level gaming motherboards (\$\$\$)
- **Creator** - designed for content creators (\$\$\$\$) -
- **Pro** - productivity and business - entry level (\$\$)



- **MEG** (Enthusiast Gaming) - high-end gaming motherboards (\$\$\$\$\$)
- **MPG** (Performance Gaming) - mid-level gaming motherboards (\$\$\$\$)
- **Creator** - designed for content creators (\$\$\$\$)
- **MAG** (Arsenal Gaming) - entry level gaming motherboards (\$\$\$)
- **Pro Series** - productivity and business motherboards (\$\$)



- **Creator** - designed for content creators (\$\$\$\$\$)
- **Taichi** - high-end gaming motherboards (\$\$\$\$)
- **Phantom Gaming** - mid-level gaming (\$\$\$)
- **Steel Legend** - entry-level gaming (\$\$)
- **Pro** - productivity and business, entry level (\$\$)

There are others....



Motherboard Sizes

8xRAM, 4xPCIe x16



E-ATX

EATX

302x330 mm

4xRAM Slots, 4xPCIe (2x16)



Standard-ATX

ATX

305x244 mm

4xRAM Slots, 3xPCIe (2x16)



Micro-ATX

mATX

244x244 mm

2 RAM Slots, 1x PCIe



ITX

Mini-ITX

170x170 mm



Pen [for scale]

CPU Sockets and Chipsets



Sponsored ⓘ

ASUS

TUF Gaming Z790-Plus WiFi LGA 1700(Intel 14th,12th &13th Gen) ATX Gaming Motherboard(PCIe 5.0,DDR5,4xM.2 Slots,16+1 DrMOS,WiFi 6,2.5Gb LAN,Front USB 3.2 Gen 2 Type-C,Thunderbolt...

★★★★★ ~ 1,256

1K+ bought in past month

\$229⁹⁹ List: \$249.99

Delivery **Thu, Feb 29**

Ships to United Kingdom

Add to cart

MSI MAG B650 TOMAHAWK WIFI ATX Motherboard

★★★★★

feefo

Read 12 reviews

- Socket AM5
- Supports 7000 Series
- DDR5 Support
- ATX Form Factor

QuickFind: 1540670

Mfr part code: MAG B650 TOMAHAWK WIFI



Sockets



You must pick the socket matching your CPU

Socket	CPU Support
AM5	AMD Ryzen 7000 series
LGA 1700	Intel Raptor Lake Refresh (14th gen) Intel Raptor Lake (13th gen) Intel Alder Lake (12th gen)
LGA 1200	Intel Rocket Lake (11th gen) Intel Comet Lake (10th gen)
AM4	AMD Ryzen 5000 Series AMD Ryzen 4000 Series AMD Ryzen 3000 Series

Chipsets

Higher numbered chipsets, higher price, more features on the mobo
Performance gains? Not so much....

Socket	Chipsets	CPU Support
AM5	X670 B650 A620	7000 Series - Ryzen 5, 7, 9
AM4	X570 B550 A520	5000 Series - Ryzen 3, 5, 7, 9
AM4	B450 A320	3000 Series - Ryzen 3, 5, 7, 9
LGA 1700	Z790 H770 B760	Intel 13th Generation Core i3, i5, i7, i9
LGA 1700	Z690 H670 B660	Intel 12th Generation Core i3, i5, i7, i9
LGA 1200	Z590 H570 B560	Intel 11th Generation Core i5, i7, i9
LGA 1200	Z490 H470 B460	Intel 10th Generation Core i3, i5, i7, i9

Does a newer CPU work with an older chipset?

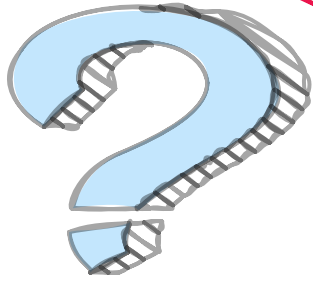
i.e. Intel 14th gen with Z690?

Yes, sometimes with a BIOS update

Does an older CPU work with a newer chipset?

i.e. Intel 12th gen with Z790?

Yes, previous gen in same socket is usually supported by default



Overall Pick 



Sponsored 

ASUS

TUF Gaming Z790-Plus WiFi LGA 1700(Intel 14th,12th &13th Gen) ATX Gaming Motherboard(PCIe 5.0,DDR5,4xM.2 Slots,16+1 DrMOS,WiFi 6,2.5Gb LAN,Front USB 3.2 Gen 2 Type-C,Thunderbolt...

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Mfr part code: MAG B650 TOMAHAWK WIFI



ASUS

TUF Gaming Z790-Plus WiFi D4 LGA 1700(Intel 14th,12th&13th Gen) ATX Gaming Motherboard(PCIe 5.0, DDR4,4xM.2 Slots,16+1 DrMOS, WiFi 6,2.5Gb LAN, Front USB 3.2 Gen 2 Type-C, Thunderbolt...

★★★★★  ~ 1,256

100+ bought in past month

\$199⁹⁹ List: \$229.99

\$14.09 delivery

Ships to United Kingdom

Add to cart

More Buying Choices

\$157.96 (22 used & new offers)

RAM Support

You must get the correct type of RAM for your motherboard

DDR types 3, 4, 5 are NOT interchangeable



PCIe Version



Sponsored ⓘ

ASUS

TUF Gaming Z790-Plus WiFi LGA 1700(Intel 14th,12th &13th Gen) ATX Gaming Motherboard(PCIe 5.0,DDR5,4xM.2 Slots,16+1 DrMOS,WiFi 6,2.5Gb LAN,Front USB 3.2 Gen 2 Type-C,Thunderbolt...

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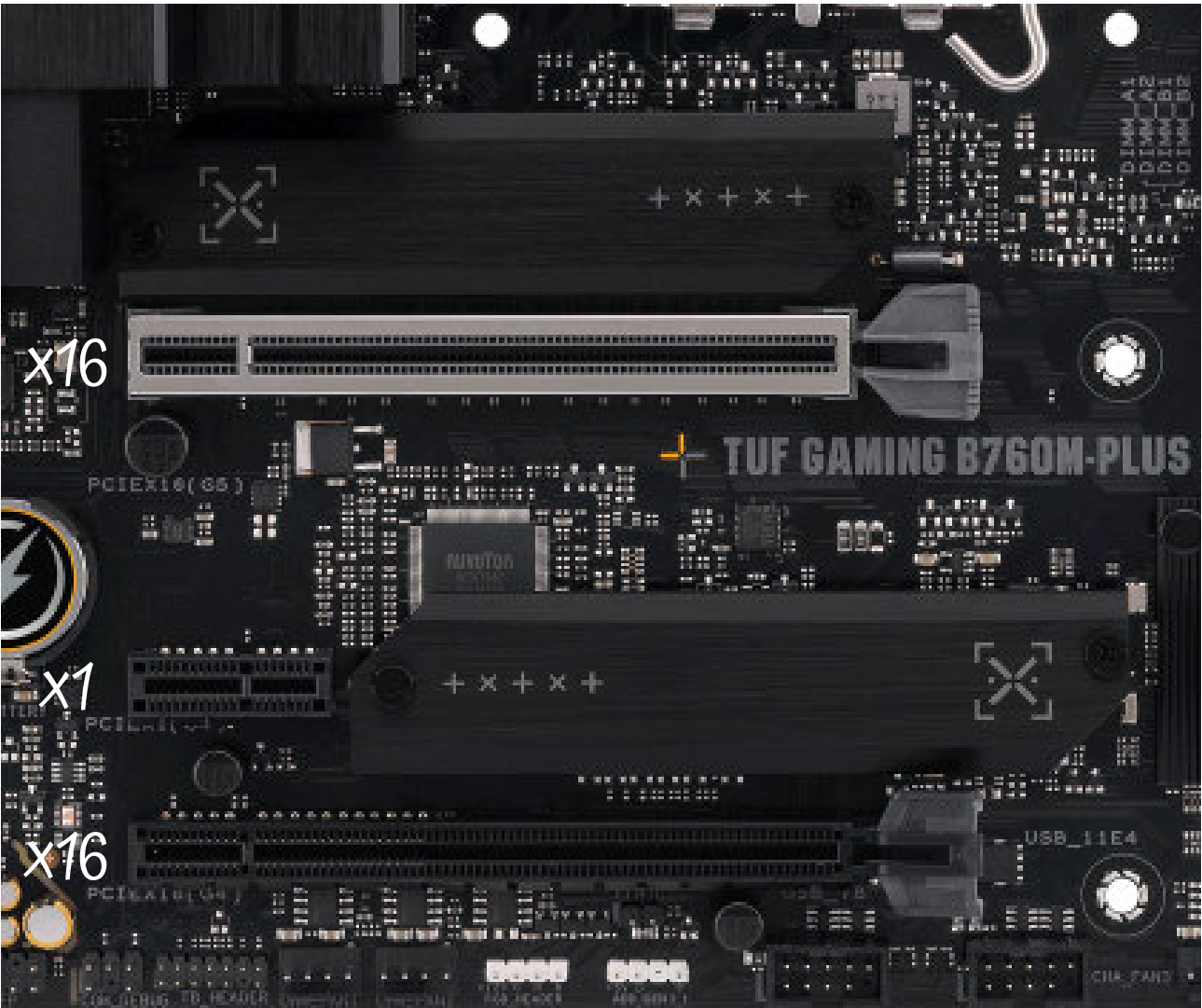
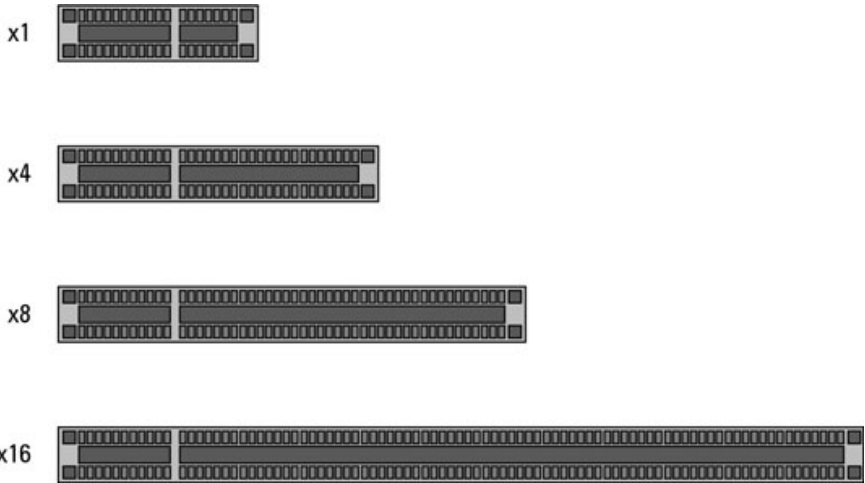
List: \$249.99

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PCIe slots are differing sizes (and speeds)
PCIe x1 cards and fit into all slots....
GPU's need x16



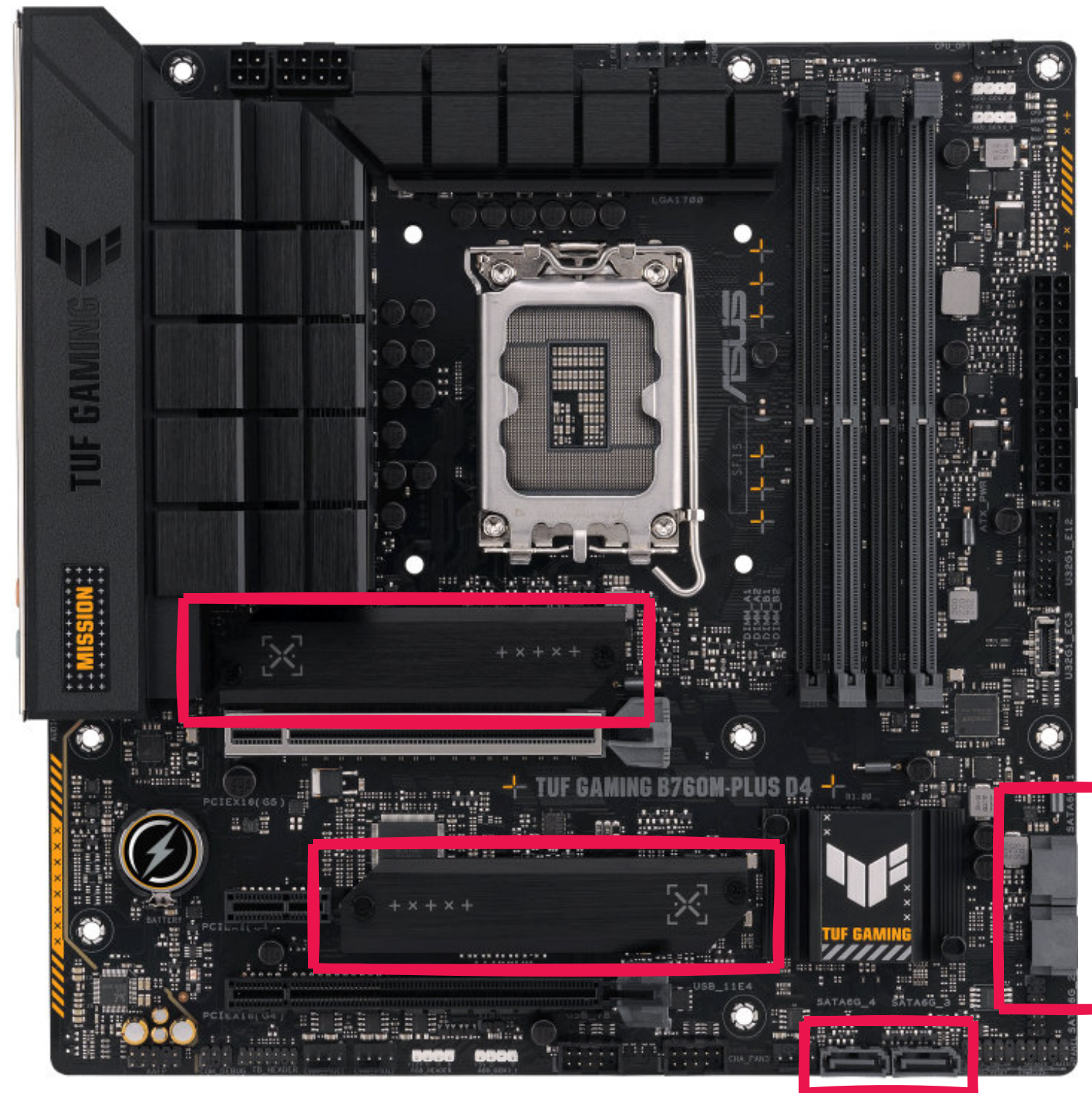
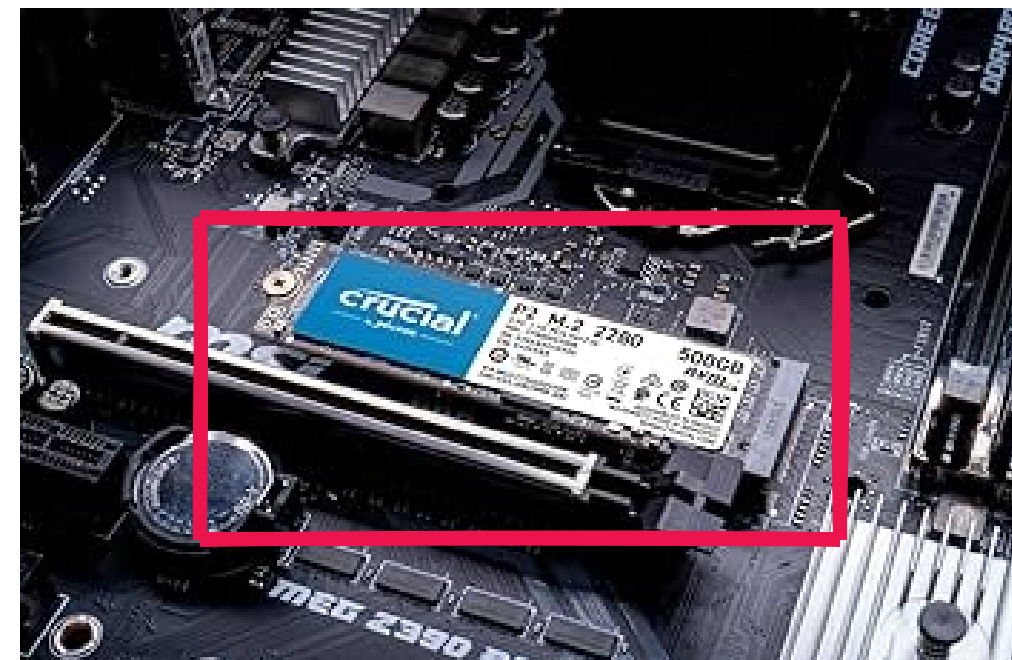
PCIe Version	Introduced	Slot Size & Throughput Speeds				
		x1	x2	x4	x8	x16
1	2003	0.25 GB/s	0.5 GB/s	1 GB/s	2 GB/s	4 GB/s
2	2007	0.5 GB/s	1 GB/s	2 GB/s	4 GB/s	8 GB/s
3	2010	0.985 GB/s	1.969 GB/s	3.938 GB/s	7.877 GB/s	15.754 GB/s
4	2017	1.969 GB/s	3.938 GB/s	7.877 GB/s	15.754 GB/s	31.508 GB/s
5	2019	3.938 GB/s	7.877 GB/s	15.754 GB/s	31.508 GB/s	63.015 GB/s
6	2022	7.563 GB/s	15.125 GB/s	30.25 GB/s	60.5 GB/s	121 GB/s
7	2025 (planned)	15.125 GB/s	30.25 GB/s	60.5 GB/s	121 GB/s	242 GB/s

Storage Connectivity

M.2 Slots

Check your M.2 fits the board

Does the board have M.2 heatsinks?



SATA Connectors

L-shape 

Usually 4x or 6x, on the side or top

For SSD's, HDD's

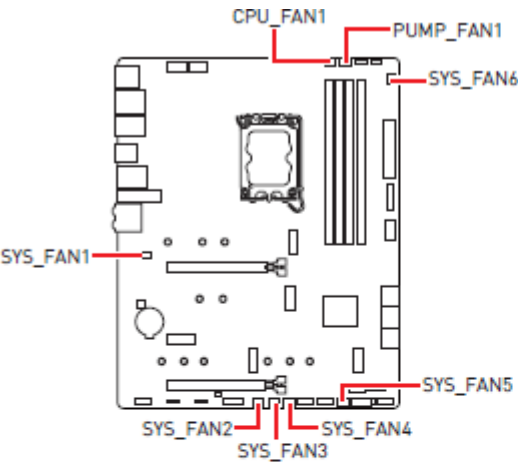
For Optical Drives (CD/DVD)



Other features to look out for...

FAN Headers 3 / 4 pin

Can you fully support your case fans and cooler?
Check the manual



1

PWM Mode pin definition

Pin	Signal Name	Pin	Signal Name
1	Ground	2	+12V
3	Sense	4	Speed Control Signal

DC Mode pin definition

Pin	Signal Name	Pin	Signal Name
1	Ground	2	Voltage Control
3	Sense	4	NC

Fan connector specifications

Connector	Default fan mode	Max. current	Max. power
CPU_FAN1	Auto mode	2A	24W
PUMP_FAN1	PWM mode	3A	36W
SYS_FAN1-6	DC mode	1A	12W



RAID

If you want data security....
Does the board allow RAID 0, 1 for your drives?
M.2 or SATA

Connectivity Options

Wi-Fi, Bluetooth, LAN Speed

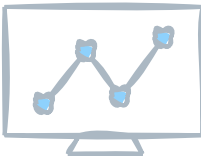


Audio Options

5.1 or 7.1? Mic? Optical?

BIOS Niceties

Fan Control
Quick flash (for overclockers)





Mobo Recommendations

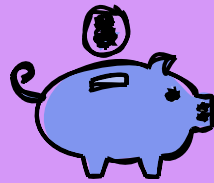
In general go with the range for your use case and your budget unless you have specific needs

Budget/Casual

Budget mobo ranges....

Less features for less price

Budget models do not run slower!



WFH Pro



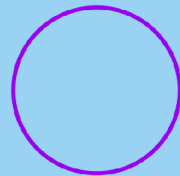
Business ranges, cheaper than gaming
no need for fancy heatsinks or LED's

Gamer

Average gamer, low to mid range gaming range will do

Adds heatsinks, look nicer

Only go high end if you really need those cutting edge features



Creative Pro / Engineer



Creator Mobos for Pro CPU's like
Threadripper (if necessary)